

Development of a Virtual Reality Assessment of Visuospatial Function and Oculomotor Control

Garima Adlakha*, Sanya Singh, Apoorva Patil, Kranthi Nuthalapati, Prajakta Khandve, Pushpak Bhattacharyya, Saravanan Manoharan, Sanjay Mallasamudram Santhanam, Isaiah J Lachica, James M Finley, Vangelis Lympouridis
University of Southern California, Los Angeles, CA, USA

ABSTRACT

This demo uses Virtual Reality (VR) to assess cognitive function in people with Parkinson's disease. We developed a VR-based assessment that combines simple game mechanics with components of the Trail Making Test. We collect performance metrics and gaze analytics during gameplay using the HTC Vive Pro Eye system. Ultimately, this data will allow clinicians and researchers to characterize cognitive and visuomotor deficits in people with neurological impairments such as Parkinson's disease.

Index Terms: Virtual Reality (VR), Cognitive Function, Parkinson's Disease, Trail Making Test, HTC Vive Pro Eye

1 INTRODUCTION

Parkinson's disease (PD) is a progressive neurodegenerative disease that affects motor function and leads to complications such as stiffness and difficulty with balance, coordination, and walking. PD can also lead to cognitive impairments that reduce the quality of life and as a result, cognitive assessments are important for guiding interventions to improve function in these individuals. One of the most common cognitive assessments is the Trail Making Test (TMT) [1]. In part B of the TMT (TMT-B), participants draw lines alternating between numbers and letters (i.e. 1-A-2-B-...-13-L) and successful performance requires attentional control, visual search, visuospatial function, planning, and task switching proficiency. Although performance in TMT-B differs between people with PD and healthy age-matched controls [2], its simple design raises questions about its ecological validity [3] and its ability to understand how distractors in the environment influence attentional control.

Attempts have been made recently to increase the ecological validity of the TMT by adapting it into a three-dimensional virtual reality (VR) environment [4]. However, questions about the generalizability of the results from this VR-based TMT to everyday behavior still remain as the VR-based task did not simulate the features of real-world environments. In addition, both the paper-based and VR-based versions of the TMT only measure time to task completion. They do not provide information on attentional control or visual search behavior, and these are critical aspects of executive function.

To overcome these limitations, we have developed a VR application following the principles of the TMT-B that consists of an immersive 3D environment with animals, 3D letters, and other objects. Users are asked to alternate between pointing at the various animals and the first letter of their name from the 3D letters located in the environment in a particular sequence following a simple game mechanic. We assess accuracy of target selection based on the sequence of the target objects hit and we also collect synchronized gaze and head orientation to objectively quantify attentional control

and visual search behavior throughout the assessment. This is critical as eye-tracking measurements, when coupled with cognitive assessments, may be able to better predict disease progression and cognitive impairment in people with PD [5].

2 METHOD

2.1 Achelois - Android platform app

We designed an Android app, Achelois, to enable clinicians to gather patients' demographics, easily access the VR game, and adjust game settings. For the demo purposes, all data are collected locally but in the future, they can be linked to a Health Insurance Portability and Accountability Act (HIPAA) compliant database or directly to the patient's health record. After entering the patient's details, the clinician can access the VR game from the menu screen where they can choose from the four available options: Tutorial, Practice, Gameplay, and Quit (Figure 1).

The 'Tutorial' option links to a video demonstrating the game structure to the patient. The 'Practice' option enables the patient to follow step-by-step instructions in the VR environment before playing the actual game. After the practice session is complete, the clinician can select the 'Gameplay' option and enter the level and complexity of the game. The corresponding scene is loaded, and the patient can experience and play the game in the VR environment. When the 'Quit' option is selected, the game ends, and all of the patient's data are saved in a CSV file for the clinician's records.

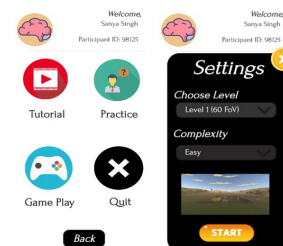


Figure 1: Achelois - Android Application

We used Photon Unity Networking (PUN) 2, a Unity package to exchange data over the network (Figure 2). The fast and reliable communication is done through dedicated Photon server(s), so clients don't need to connect one to one. When the game is launched, users are connected to the Photon Unity Networking (PUN) server. When the clinician chooses Practice, Gameplay, or Quit, the chosen options are sent to the VR app via Photon cloud.

2.2 Game Design

The game involves selecting a series of targets while alternating between the letters and animal names starting with these letters. For example, the patient needs to select the letter A followed by ant, B followed by a butterfly, and so on. This game challenges the patient's ability to visually scan the environment, identify targets, and practice inhibitory control by being able to transition between

*e-mail: gadlakha@usc.edu

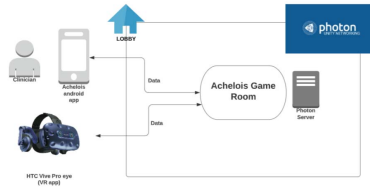


Figure 2: Communication between android APP and VR APP

the letters and animal names.

There are three levels and each level has three modes of complexity. A level is set to be complete once the patient selects the target letters and the animals in the correct sequential order. Level 1 involves the patient selecting four letters and four animals. In this level, the targets are placed within a 60 degree Field of View (FoV). Similarly, level 2 has six letters and six animals with FoV set to 120 degrees. Level 3 has eight letters and eight animals with the FoV set to 210 degrees.

The three complexities involved at each level are easy, medium, and difficult. The game with 'easy' complexity has the targets placed without any distractions in the environment. The game with 'medium' complexity has targets placed with minimal distractions. Lastly, the game with 'difficult' complexity has targets placed with maximum background distractions.

The game's algorithm is designed in such a way that when the patient selects the intended target, a 'Correct' message is displayed. Similarly, when the patient selects an incorrect target, a 'Try Again' message is displayed and the patient has to click on the correct target to proceed. Finally, when the patient has completed the game, a 'Congratulations' message is displayed. We also implemented a target layout algorithm that produces different levels by varying the location of targets even if the clinician loads the same level and complexity. Furthermore, the entire sequence of the patient's gameplay is recorded to calculate the accuracy in identifying the targets.

2.3 Eye Tracking

People with PD have difficulties performing saccades (rapid eye movement) and smooth pursuits (following a target). Eye tremors in PD patients can be detected early on even before the onset of cardinal symptoms like postural instability (balance problems), bradykinesia (slow movements), amnesia, etc. To assess a PD patient's cognitive and visuospatial function with the help of these early signs, we decided to track every eye and head movement using the HTC Vive Pro Eye. The gaze data are sampled at a frequency of 120 Hz, with a trackable FoV of 110 degrees and the interface used is HTC SRanipal SDK in Unity. The script captures six key eye variables: eye openness (a number ranging from 0 to 1 with 0 being closed and 1 being open), gaze origin (a 3D vector capturing the origin of the gaze), gaze direction (a 3D vector indicating the direction in which the patient is looking), pupil diameter (diameter of the pupils in mm), pupil position (a 2D vector indicating the position of the pupil in the virtual space), convergence distance (distance between the pupils in mm), two head movement variables: head position (a 3D vector indicating the absolute position of Head Mounted Display in the virtual space) and head orientation as a quaternion, and a precise timestamp.

3 RESULTS

Once a level is completed, a CSV file is generated which includes the patient's details, correct sequence, the patient's sequence, and the score. The score is calculated by comparing the correct sequence and the sequence taken by the patient. Another CSV file is generated containing the patient's eye and head movement data.

4 DEMO

Users can experience the immersive environment using the HTC Vive Pro Eye and clinicians can interface with the VR environment and its settings using an Android application. The virtual environment was developed in Unity 2019.4.6. The primary focus of our research is to gather the eye data and head movements of people with PD to assess their cognitive function using a VR game inspired by the TMT. The demo will start by collecting the demographics of the user in the Android application. To make users familiar with the assessment, we have developed an option "Video Tutorial" where they are required to watch a video and have a basic understanding of the assessment structure. This can help them learn the interface and gameplay. There are instructions alongside the game so that the users can fully understand the game's objective. After following the instructions and completing the tutorial, the user can play the game. Clinicians can select Gameplay mode and enter the level and complexity of the game. After they select the Quit button, their data will be saved in CSV files and the game will end.



Figure 3: Achelois - VR Environment

5 FUTURE WORK

In future work, we will store the patient's demographics and eye-related data in a HIPAA compliant cloud database for clinical use. Further, we will compare the trajectory of eye movements from a set of age-matched controls to the eye movements of patients during gameplay to understand how PD affects visuomotor processing. Heat-maps can be created to visualize data on the eye and head movements. As a future enhancement, we can provide login security for the mobile app, increase the intensity of the game, and add more visual-cognitive games. We will also validate the application by comparing users' performance to their performance in the paper-based version of the TMT. In the future, this research work can be used to potentially diagnose cognitive impairments. This might also form the basis of a training intervention later on.

REFERENCES

- [1] T. A. Salthouse, "What cognitive abilities are involved in trail-making performance?" *Intelligence*, vol. 39, no. 4, pp. 222 – 232, 2011.
- [2] M. Rozenfeld Olchik, M. Ghisi, A. Freiry, A. Ayres, A. Schumacher Schuh, C. Rieder, and A. Teixeira, "Comparison trail making test between individuals with parkinson's disease and health controls: Suggestions of cutoff point?" *Psychology Neuroscience*, vol. 10, pp. 77–82, 03 2017.
- [3] M. Plotnik, G. M. Doniger, Y. Bahat, A. Gottlieb, O. Ben Gal, E. Arad, L. Kribus-Shmiel, S. Kimel-Naor, G. Zeilig, M. Schnaider-Beeri, R. Yanovich, I. Ketko, and Y. Heled, "Immersive trail making: Construct validity of an ecological neuropsychological test," in *2017 International Conference on Virtual Rehabilitation (ICVR)*, 2017, pp. 1–6.
- [4] O. Ben Gal, G. M. Doniger, M. Cohen, Y. Bahat, M. S. Beeri, G. Zeilig, and M. Plotnik, "Cognitive-motor interaction during virtual reality trail making," in *2019 International Conference on Virtual Rehabilitation (ICVR)*, 2019, pp. 1–6.
- [5] M. K. Eckstein, B. Guerra-Carrillo, A. T. Miller Singley, and S. A. Bunge, "Beyond eye gaze: What else can eyetracking reveal about cognition and cognitive development?" *Developmental Cognitive Neuroscience*, vol. 25, pp. 69 – 91, 2017.